Portable fire extinguishers apply an extinguishing agent that will either cool burning fuel, displace or remove oxygen, or stop the chemical reaction so fire cannot continue to burn. When the candle of an extinguisher is compressed, it opens and inner canister of high pressure gases forces the extinguishing agent from the main cylinder through a siphon tube and out the nozzle. A fire extinguisher works much like a can of hair spray.

Types of fire extinguisher:

The most common types of fire extinguishers are,

- ❖ Air pressurized water extinguishers,
- ❖ Carbon-di-oxide extinguishers.
- ❖ Dry chemical powder extinguishers.

The different types of extinguisher tackle different types of fire

Type	Class A Combustible materials (e.g. paper & wood)	Class B Flammable Liquids (e.g. paint & petrol)	Class C Flammable Gases (e.g. butane & methane)	Class D Flammable metals (e.g. lithium & potassium)	Electrical Electrical Equipment (e.g. computer & generators)	Class F Deep fat fryers (e.g. chip pans)	Comments
Water	✓	✗	✗	✗	✗	✗	Do not use on liquid or electric fires
Foam	✓	✓	✗	✗	✗	✗	Not suited to domestic use
Dry powder	✓	✓	✓	✓	✓	✗	Can be used safely up to 1000 volts
CO_2	✗	✓	✗	✗	✓	✗	Safe on both high and low voltage
Wet chemical	✓	✗	✗	✗	✗	✓	Use on extremely high tempertures

Fire extinguishers can be broadly classified into five types:

1. Water, 2.Foam, 3.Dry Powder, 4.CO_2, 5.Wet Chemical.

The classes of fire

There are five classes of fire: Class A, Class B, Class C, Class D, and Class E.

Class A fires	–	**Combustible materials:** caused by flammable solids, such as wood, paper, and fabric.
Class B fires	–	**Flammable liquids:** such as petrol, turpentine or paint.
Class C fires	–	**Flammable gases:** like hydrogen, butane or methane.
Class D fires	–	**Combustible metals:** chemicals such as magnesium, aluminum or potassium.
Class E fires	–	Typically a chip-pan fire.
Electrical fires	–	**Electrical equipment:** once the electrical item is removed, the fire changes class.

VII STD Science Term-3 EM unit 4.indd 77 06-09-2022 15:23:08